

# Seeing Images as Graphs for Structured Visual Representation Learning

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## Abstract

Vision Transformers (ViTs) achieve strong performance by processing images as unstructured sequences of patches, but this flat view discards the rich spatial topology and content-driven groupings that are intrinsic to 2D visual data. Existing remedies either impose rigid locality constraints or entangle structural and semantic signals into opaque, monolithic mechanisms. We argue that the fundamental limitation is one of perspective: images are better understood not as bags of tokens, but as **two complementary graphs**—a fixed lattice graph encoding geometric relationships and patch roles, and a dynamic contextual graph encoding pure visual similarity. In this paper, we introduce the **Semantic Vision Graphormer (SVG)**, a ViT architecture that explicitly injects graph-theoretic inductive biases from both graphs into self-attention. Three lightweight, interpretable bias terms are added to attention scores: (i) a spatial bias from shortest-path distances on the lattice; (ii) a centrality encoding via learnable degree-based scalars; and (iii) a semantic bias computed as bucketed cosine similarity of content-only patch embeddings, kept cleanly separated from positional augmentation. Layer-adaptive scalars allow each transformer layer to dynamically balance geometric and semantic priors. SVG adds fewer than 15k parameters and requires negligible MAC(Multiply-Accumulate operations) overhead. On CIFAR-10, it outperforms ViT and Swin baselines by **2.18%** while matching their parameter count (5.4M), demonstrating that explicitly modeling images as dual graphs significantly enhances representation learning.

## 1 Introduction

Since the introduction of the Vision Transformer (ViT), the field of computer vision has witnessed a paradigm shift: global self-attention over image patches can rival and even surpass convolutional networks, provided sufficient data and scale. In its original form, a ViT slices an image into a grid of non-overlapping patches, linearly projects each patch into a token, adds a positional encoding, and lets a cascade of transformer layers model interactions among all tokens. This design is elegant and scalable, but it harbors a deep conceptual tension: the image is presented to the model as a *set* of tokens, yet it is inherently a *spatially organized structure* with well-defined topology and rich contextual groupings.

The standard ViT addresses the lack of spatial awareness only indirectly. Absolute positional encodings inform a token about *its own* location, but they do not encode the *relations* between locations—how far apart two patches are, whether a patch occupies a corner, edge, or interior, or whether two visually identical patches belong to the same object. Self-attention must infer all such regularities purely from data, a task that is sample-inefficient and consumes substantial model capacity, especially on small- and medium-sized datasets.

Subsequent variants have attempted to recover structural sensitivity. The Swin Transformer reintroduces hard locality via windowed attention, while ConViT blends convolutions with self-attention. However, these strategies fuse spatial and content information inside opaque mechanisms, making it difficult to disentangle "where" a patch is from "what" it depicts. Other approaches have turned to explicit graph representations, such as Vision GNN (ViG) or Graph Vision Transformers (GVT). While conceptually appealing, these models dynamically rewire  $k$ -NN graphs at every layer, resulting in  $\mathcal{O}(N^2D)$  distance computations and sparse message passing. While this is similar to the self-attention computation complexity, because modern GPUs are heavily optimized for dense matrix multiplications, these sparse graph operations suffer from severe hardware underutilization, often making them slower in practice than dense ViTs.

In this work, we propose a fundamental change of perspective. Instead of treating an image as a flat collection of tokens, **we view it as a graph**—in fact, as two graphs at once. Every image naturally gives rise to:

- A **lattice graph**: a fixed, 8-connected grid whose nodes are patches and whose edges encode immediate spatial adjacency. From this graph one can derive purely structural properties: the shortest-path distance between any two patches, and the degree (centrality) of each patch: corner (degree 3), edge (degree 5), or interior (degree 8). This graph captures the skeleton of the image: *how far apart* things are, and *how structurally important* a patch is.
- A **contextual graph**: a dynamic, image-dependent graph whose edges connect patches that are visually similar. This graph captures the flesh: *what* a patch contains and *which patches belong together semantically*, regardless of their spatial separation.

These two graphs are orthogonal and complementary. The lattice graph offers a fixed spatial coordinate system and a notion of structural role; the contextual graph offers a content-driven grouping that reflects object parts, textures, and visual concepts. A model that can consult both graphs simultaneously would know not only *that two patches are close*, but also *that they look alike*, and could use these signals to guide attention without having to rediscover geometry or semantics from scratch.

Our key insight is to make this dual-graph structure explicit, and to inject the resulting inductive biases directly into the self-attention computation of a ViT. This idea is inspired by the Graphormer architecture for graph representation learning, which demonstrated that Transformers can master graph tasks when supplied with degree centrality and shortest-path spatial biases. We adapt and extend Graphormer’s philosophy to vision, introducing three lightweight, interpretable bias terms that are added to the pre-softmax attention logits:

1. **Spatial bias** from the lattice graph: a learnable scalar for each possible shortest-path distance between patches, plus a dedicated bias for the virtual connection to the [CLS] token.
2. **Centrality encoding** from the lattice graph: a single learnable scalar per degree class, concatenated to each patch embedding as a dedicated structural feature dimension.
3. **Semantic bias** from the contextual graph: a **bucketed cosine similarity** computed on the raw, content-only patch embeddings *before* any positional or centrality augmentation. This guarantees that the similarity signal reflects pure visual content, untainted by spatial information. Similarities are quantized into coarse bins (low, medium, high, self) then mapped to learnable bias values.

Why add an explicit semantic bias when self-attention itself performs content-based comparison via query-key dot products? The answer lies in the clarity of the signal. Standard QK similarity operates on tokens that already carry positional and centrality encodings; thus, two identical patches in different locations will have different Q and K vectors, and their dot product conflates content similarity with positional disagreement. The explicit cosine bias, computed on untouched patch projections, provides a pure, position-independent signal that acts as a robust prior and reduces the burden on the learned attention mechanism. The model remains free to override or amplify this bias via the QK term, but it no longer needs to learn to factor out position from scratch.

All biases are precomputed per batch, and shared across attention heads. Crucially, we introduce layer-adaptive learnable scalars for the spatial and semantic biases, allowing each transformer layer to independently adjust its reliance on geometric versus content priors. Early layers may lean more on spatial locality to build low-level features, while deeper layers may shift toward semantic groupings for object-level reasoning.

The resulting architecture, which we call the **Semantic Vision Graphormer (SVG)**, embodies a concrete instantiation of the dual-graph philosophy. It adds fewer than 15k parameters to a standard ViT and is computationally indistinguishable from it at inference. On CIFAR-10, SVG achieves **94.68%** test accuracy, outperforming ViT-timm by 4% (*subject to change*) and Swin-timm by 2.18% (*subject to change*), while using only 5.4 million parameters.

From an inductive bias perspective, SVG reduces the sample complexity of visual representation learning by explicitly factoring out geometric and semantic regularities that standard ViTs must infer from data alone.

By injecting graph-theoretic priors directly into attention logits, we constrain the attention distribution toward structurally and semantically coherent neighborhoods, effectively shrinking the hypothesis space. Layer-adaptive scalars further enable the model to modulate prior strength hierarchically, aligning with known principles of progressive feature abstraction in deep networks.

We stress that our contribution is not simply a new set of architectural tweaks. It is a **shift in how we think about images for transformers**. By recognizing the dual graph nature of images and explicitly encoding both their structural skeleton and their semantic flesh, we provide a modular, interpretable, and statistically efficient framework that eases the learning problem and opens the door to a new family of graph-informed vision models.

The rest of this paper is organized as follows. Section 2 discusses related work, situating SVG with respect to graph-informed vision and transformer architectures. Section 3 presents the dual-graph perspective and the methods of the SVG design. Section 4 reports experimental results, and Section 5 concludes with a discussion of limitations and future directions.

## 2 Related Work

Our work builds upon and synthesizes ideas from four primary research streams: (1) Vision Transformers and the reintroduction of structural priors, (2) Efficient convolutional architectures, (3) Graph-based representations in vision, and (4) the injection of structural biases into self-attention. We discuss each in turn, highlighting the limitations that motivate our dual-graph approach.

### 2.1 Vision Transformers and Structural Priors

The Vision Transformer (ViT) established that global self-attention over image patches can rival convolutional networks, but it treats images as unstructured 1D sequences, discarding 2D spatial topology. This leads to sample inefficiency, particularly on smaller datasets like CIFAR-10. Subsequent works have attempted to recover structural sensitivity through various means:

- **Hard Locality Priors:** The Swin Transformer reintroduces locality by restricting self-attention to non-overlapping windows, with shifted windows for cross-window connection. While this improves efficiency, it sacrifices a true global receptive field within layers and imposes a rigid, human-specified neighborhood structure.
- **Soft Locality Priors:** ConViT learns a gated mixture of convolutional and content-based attention, while CvT incorporates convolutional projections into the attention mechanism. While these methods successfully blend spatial priors with attention, they entangle spatial and semantic information within a single, opaque mechanism, making it difficult for the model to disentangle "where" a patch is from "what" it depicts.

Our approach, SVG, differs fundamentally. Rather than modifying the attention mechanism itself or imposing rigid windows, we inject lightweight, explicit graph-theoretic biases directly into the attention logits. This preserves the global receptive field of standard ViT while cleanly separating spatial structure from semantic content.

### 2.2 Efficient Architectures and Convolutional Priors

To contextualize the efficiency of our method, we also consider strong convolutional baselines. ResNet established the standard for translationally-invariant hierarchical feature extraction via local convolutions. More recently, ConvMixer demonstrated that pure depthwise convolutions for patch mixing can achieve impressive accuracy with minimal parameters, challenging the necessity of self-attention entirely.

While these models possess strong inductive biases for local features, their rigid sliding-window nature limits their ability to capture long-range dependencies without deeply stacking layers. SVG matches the lightweight parameter footprint of models like ConvMixer but retains the global context capability of Transformers, augmented by flexible graph priors rather than rigid convolutional kernels.

## 2.3 Graph-Based Representations in Vision

Representing images as graphs offers a flexible alternative to rigid grids or unstructured sequences, but prior approaches suffer from severe representational and hardware-level bottlenecks.

Historically, **superpixel-based GCNs** constructed graphs where nodes represent perceptually similar pixel groupings. While this captures a primitive form of content-driven grouping, it suffers from fundamental limitations: the hard assignment of pixels to superpixels creates rigid boundary bottlenecks, pooling destroys fine-grained intra-superpixel variance, and the superpixel generation algorithms (e.g., SLIC) are non-differentiable. This decouples the graph topology from the end-to-end loss, preventing the feature extractor from optimizing for the downstream graph task.

To capture more flexible semantics, recent works have turned to dynamic graph construction in feature space. **DGCNN** introduced EdgeConv, dynamically connecting nodes to their  $k$ -nearest neighbors. While effective, DGCNN treats the input purely as a set of vectors, lacking explicit geometric priors and entirely ignoring the underlying spatial lattice. In the vision domain, models like **Vision GNN (ViG)** and various **Graph Vision Transformers (GVT)** apply similar dynamic  $k$ -NN rewiring to image patches. However, these approaches introduce severe computational and hardware bottlenecks. Computing pairwise distances for  $k$ -NN search at every layer is  $\mathcal{O}(N^2D)$  and relies on irregular, sparse message passing or sparse attention. Because modern GPUs are heavily optimized for dense, highly parallel matrix multiplications (e.g., via FlashAttention), these sparse graph operations suffer from poor memory access patterns and low hardware utilization, often making them slower in practice than dense ViTs. Furthermore, deep dynamic rewiring makes these models highly susceptible to over-smoothing, where node features become indistinguishable and spatial structure is destroyed.

Our dual-graph perspective circumvents these pitfalls. Unlike superpixel GCNs, SVG is fully differentiable and preserves fine-grained patch details. Unlike ViG or GVT, we do not replace self-attention with computationally expensive  $k$ -NN graph convolutions or sparse attention. Instead, we recognize that every image intrinsically yields two orthogonal graphs: a fixed **lattice graph** (providing the explicit geometric priors missing in DGCNN) and a dynamic **contextual graph**. We inject lightweight, dense biases derived from both graphs directly into the Transformer’s self-attention logits. This preserves the global, hardware-efficient dense context of standard ViTs while adding rich structural and semantic awareness with negligible MAC(Multiply-Accumulate operations) overhead.

## 2.4 Structural Biases in Self-Attention

A key line of research modifies the self-attention computation itself to encode structural information. Early works on **Relative Position Representations (RPR)** introduced additive biases to attention logits based on relative distances, a concept later popularized in language models by methods like **ALiBi** for length extrapolation. While successful in 1D sequences, these approaches are limited to fixed, predefined notions of distance and do not capture content-driven semantic groupings in 2D spaces.

Most relevant to our work is **Graphormer**, which demonstrated that Transformers can excel at graph tasks when supplied with explicit graph-theoretic inductive biases. It introduces Spatial Encoding (based on shortest-path distances) as an additive bias to attention logits. However, Graphormer was designed for irregular graph data (e.g., molecules) and relies purely on fixed graph topology, lacking a mechanism for dynamic, content-driven relationships.

Our Semantic Vision Graphormer (SVG) extends the Graphormer philosophy to vision by introducing a **third bias**: a dynamic **Semantic Bias** computed from bucketed cosine similarity of content-only patch embeddings. This creates the first dual-graph Transformer architecture for images. Crucially, we compute semantic similarity on raw patch projections *before* adding positional or centrality information, ensuring the signal remains pure. We also introduce **layer-adaptive scalars**, allowing each layer to independently balance geometric and semantic priors—early layers may favor spatial locality for low-level features, while deeper layers shift toward semantic grouping for object reasoning. This addresses a limitation of prior methods like Graphormer, which apply uniform biases across all layers.

## 2.5 Discussion and Positioning

In summary, prior works have incorporated structural priors into vision models in a fragmented or rigid manner. Swin forces hard locality, ConViT entangles signals, Superpixel GCNs suffer from non-differentiable hard assignments, and dynamic graph models like ViG introduce severe hardware inefficiencies via sparse operations. Our work proposes a fundamental shift: **images are naturally dual graphs**. By explicitly injecting both structural and semantic biases as lightweight, interpretable, and hardware-friendly dense terms into self-attention, SVG provides a modular, statistically efficient, and computationally negligible framework that outperforms strong baselines across both Transformer and convolutional paradigms.

## 3 Method

The Semantic Vision Graphormer (SVG) enhances the Vision Transformer with structural awareness by integrating graph-theoretic inductive biases into self-attention. Unlike standard ViTs that treat patches as a flat sequence, we view the image as two complementary graphs: a fixed **lattice graph** that encodes the 2D spatial topology, and a dynamic **contextual graph** that captures content-based semantic groupings. From these two graphs we derive three explicit biases that are added directly to the attention logits: spatial distance, structural centrality, and semantic similarity.

### 3.1 Dual Graph Construction and Patch Representation

An image of size  $H \times W \times C$  is partitioned into  $N = HW/P^2$  patches of size  $P \times P$ . Each patch is flattened and projected via a linear layer:

$$x_p^{(i)} = E \cdot \text{flatten}(\text{Patch}_i(X)) \in \mathbb{R}^D, \quad (1)$$

where  $E \in \mathbb{R}^{(P^2 C) \times D}$ .

We construct two graphs over the set of patches  $V = \{1, \dots, N\}$ :

- **Lattice graph**  $\mathcal{G}_{\text{lat}} = (V, E_{\text{lat}})$ . Edges follow 8-connectivity, so that a patch’s neighbours are those touching it horizontally, vertically, or diagonally. This graph is fixed for all images and captures the spatial skeleton: patch roles (corner, edge, interior) and pairwise distances.
- **Contextual graph**  $\mathcal{G}_{\text{ctx}} = (V, E_{\text{ctx}})$ . Edges are defined implicitly by visual similarity between patches. Specifically, we compute a semantic affinity score from the raw patch embeddings  $x_p$  (before any positional or centrality augmentation) and use it to bias attention. This graph is dynamic, image-dependent, and captures the flesh of the image: which patches belong together by content.

All subsequent biases are derived from these two graphs—spatial distance and centrality from  $\mathcal{G}_{\text{lat}}$ , semantic similarity from  $\mathcal{G}_{\text{ctx}}$ .

### 3.2 Centrality Encoding (Scalar Augmentation)

Following Graphormer’s insight that node importance correlates with connectivity, we assign each patch a structural role based on its degree in  $\mathcal{G}_{\text{lat}}$ :  $d_i \in \{3, 5, 8\}$  for corner, edge, interior, respectively. Rather than a full vector embedding (which adds  $O(D)$  parameters), we use a scalar augmentation for efficiency:

$$c_i = z(d_i), \quad z : \{3, 5, 8\} \rightarrow \mathbb{R}, \quad (2)$$

where  $z$  is implemented as a learnable embedding with 3 scalars. The patch embedding is augmented by concatenation:

$$\tilde{x}_p^{(i)} = [x_p^{(i)}; c_i] \in \mathbb{R}^{D+1}, \quad (3)$$

preserving centrality as a dedicated feature dimension. The [CLS] token receives a centrality of zero, signalling that it is not part of the spatial grid.

### 3.3 Structural and Semantic Attention Biases

We modify self-attention to incorporate the bias terms as shown below:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \alpha_\ell^{(\text{sp})} B^{\text{sp}} + \alpha_\ell^{(\text{sem})} C^{\text{sem}}\right) V \quad (4)$$

where  $d_k$  is the dimension of the key vectors,  $\ell$  is the layer index and  $\alpha_\ell^{(\cdot)}$  are learnable scalars (initialised to 1).

**Spatial Bias  $B^{\text{sp}}$  from the lattice graph:** Let  $\phi(i, j)$  be the shortest-path (Chebyshev) distance between patches  $i$  and  $j$  on  $\mathcal{G}_{\text{lat}}$ . We define a shared learnable scalar  $b_{\phi(i, j)} \in \mathbb{R}$  for each distance  $\phi \in \{0, 1, \dots, D_{\text{max}}\}$ , and a separate scalar  $b_{\text{virt}}$  for connections to the [CLS] token (virtual node). This follows Graphormer’s practice of distinguishing physical from virtual edges. When learned to decrease with distance, this bias naturally encourages local attention while preserving the global receptive field, or vice versa.

**Semantic Bias  $C^{\text{sem}}$  from the contextual graph:** To avoid conflating position/centrality with content, cosine similarity is computed only on the projected patches  $x_p \in \mathbb{R}^D$  (before centrality/positional augmentation). We bucket pairwise similarities:

$$\sigma(i, j) = \begin{cases} 0 & \cos < 0.4, \\ 1 & 0.4 \leq \cos < 0.7, \\ 2 & \cos \geq 0.7, \end{cases} \quad [C^{\text{sem}}]_{ij} = w_{\sigma(i, j)} \quad (i, j \geq 1), \quad (5)$$

with  $[C^{\text{sem}}]_{ij} = 0$  if  $i = 0$  or  $j = 0$  (no semantic bias for [CLS]). The self-connection ( $i = j$ ) is fixed to zero bias. Bucketing provides a robust, quantised affinity signal while preventing overfitting to small cosine fluctuations. This bias captures the edges of  $\mathcal{G}_{\text{ctx}}$ : patches that look alike are encouraged to attend to each other via shared learnable parameters, irrespective of their location in the lattice.

### 3.4 Final Representation

A [CLS] token (with zero centrality) and learnable absolute positional embeddings is added, yielding input  $X^{(0)} \in \mathbb{R}^{(N+1) \times (D+1)}$ . After  $L$  transformer layers, the [CLS] token’s final state serves as the image representation for downstream tasks.

### 3.5 Computational Complexity and Hardware Efficiency

A primary motivation for SVG is the injection of dual-graph priors without sacrificing the hardware efficiency characteristic of dense Transformers. The theoretical complexity of standard self-attention is  $\mathcal{O}(N^2 D)$ , dominated by the dense Query-Key and Attention-Value matrix multiplications. SVG preserves this exact complexity class. The spatial bias ( $B^{\text{sp}}$ ) is derived from a static lattice and requires  $\mathcal{O}(1)$  lookup operations. The semantic bias ( $C^{\text{sem}}$ ) requires an  $\mathcal{O}(N^2 D)$  matrix multiplication to compute pairwise cosine similarities; however, because this is computed on the raw, unaugmented patch embeddings, it can be precomputed once per forward pass and cached. Crucially, the injection of these biases into the attention logits requires only  $\mathcal{O}(N^2)$  element-wise additions. Unlike dynamic graph models (e.g., ViG, DGCNN) that require  $\mathcal{O}(N^2 D)$  pairwise distance calculations followed by branch-heavy  $\mathcal{O}(N^2 \log k)$  Top-K sorting and irregular sparse message passing, which are poorly suited for the Single Instruction, Multiple Threads (SIMT) architecture of GPUs; SVGs operations consist entirely of dense, highly-parallelizable tensor operations. This allows SVG to map perfectly onto modern IO-aware dense attention kernels (e.g., FlashAttention), resulting in zero measurable wall-clock latency overhead at inference time compared to the ViT baseline, despite the rich structural inductive biases.

**Key Advantages.** (1) Biases are precomputed per batch causing no runtime overhead; (2) Very few parameters added to baseline ViT (<15k); (3) Clear separation of *where* (spatial), *how important* (centrality), and *what* (semantic) signals, making the attention mechanism more interpretable.

## 4 Experiments

*Ablation study in progress.*

### 4.1 Ablation Study

**Training setup:**

Epochs: 500(approximate)

Optimizer: AdamW, lr=1e-3, weight\_decay=0.05

Batch\_size: 128

LR schedule: Cosine annealing, linear warmup 5 epochs

Data augmentation:

1. RandomCrop(32, padding=4) + RandomHorizontalFlip

2. RandAugment(n=2, m=10)

3. MixUp(=0.8) + CutMix(=1.0) (applied with equal probability)

4. RandomErasing(p=0.25)

Label smoothing: 0.1

*(Removed)Drop path: 0.1 for every layer => Caused conflicts with torch.compile()*

*(Removed)Exponential Moving Average: Decay=0.9999 => Caused extremely slow convergence*

Mixed precision: torch.bfloat16

Loss: CrossEntropy with Mixup/CutMix mixing

Table 1: CIFAR-10 results.

| Model                        | Accuracy (%) | # Epochs |
|------------------------------|--------------|----------|
| <b>SVG(Main)</b>             | 94.68        | 479      |
| SVG w/o semantic             | 94.51        | 492      |
| SVG w/o (semantic & spatial) | 92.95        | 500      |